Ahmed Khaled

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Education

M.Sc. in Informatics — **Nile University** — *Giza, Egypt* — **04/2024 - present**

- **GPA**: **4.0** / 4.0
- **Coursework**: Machine Learning, Scientific Computing, Natural Language Processing, Image Processing and Computer Graphics, Quantum Computing, Bioinformatics, Deep Learning.

Bachelor of Computer Science — **Ain Shams University** — *Cairo, Egypt* — **08/2019 - 07/2023**
Scientific Computing Department

- Ranked **FIRST** in Bachelor*
- Self-Directed Learning: OSSU Systematic Program Design, Programming Languages, Missing Semester of CS Education, Reinforcement Learning

Skills

- **Problem Solving**: Data Structures | Algorithms | C++ | python
- **Functional Programming**: sml | racket
- **Databases**: sql | mongoDB | PostgreSQL | NoSQl
- **Tools**: git & Github | bash | Docker
- **Specialized Areas**: Medical Computer Vision, Natural Language Processing
- **Languages**: Arabic (Native), English(Proficient)

Experience

Teaching Assistant (full-time) — **Nile University** — *Giza, Egypt* — **02/2024 - present!**

- Assist in teaching **Algorithms Analysis and Design for, and Deep Learning for Postgraduates.**
- Assist in teaching **Introduction to Big Data**, **Discrete Mathematics**, **Machine Intelligence, Data Structures and Algorithms, Logic Design, Databases, and Programming for Undergraduates,**

Computer Science Mentor(part-time) — **Almentor** — *online* — **12/2023 - 12/2024**

- Actively participating in the implementation of the **DECI Initiative** as presented by the Egyptian Ministry of Communications.

Coding Instructor(full-time) — **Discovery Academy** — *on-site, Fifth Statement, Egypt* — **08/2023 - 01/2024**

- Design long-term Learning Curricula composed of **10 levels** of Programming Experience.

Software Engineer, Intern — **Gensystem** — *Remote, Saudi Arabia* — **07/2023 - 12/2023**

- Contributed to the development of “Curious”, a Voting System in Saudi Arabia, using **Supabase + Flutter Flow Stack**, by designing the Database Schema efficiently, and applied Database row-level Security besides an Authentication System.

Projects

- **Master THESIS :** MRI Prognosis for Brain Metastasis.
- **Aftina:** Language Model that Leverages RAG System-based that relies on **Trusted** Fatwas Database from Dar Al Iftaa
- **SOCCER Graduation Project - Excellent***

Retrieving the Exact Time when a soccer action (goal, card, foul) occurs, on which an Automatic Highlighting Generator can be built, using context-aware loss Function (Python, PyTorch, Computer Vision). [Code] [Paper] [Documentation] [Demo]. **(07/2023)**

- **Kaggle Competition, White Blood Cell Classification Under Class Imbalance and Domain Shift.**

Retrieving the Exact Time when a soccer action (goal, card, foul) occurs, on which an Automatic Highlighting Generator can be built, using context-aware loss Function (Python, PyTorch, Computer Vision). [Code] [Paper] [Documentation] [Demo]. **(07/2023)**

- **Parallel Low Pass Filter:** Using Message Passing Model to implement Distributed Parallel Architecture, applying Convolution Operation on Decomposed Image then Integration. [code] **(05/2023)**
- **FOS:** Developing Principal Modules, Memory Management, CPU Scheduling, Queueing System, and Parallel Execution of Processes(multi-threading) to implement a functional Operating System using C language.[code] **(05/2022)**
- **MIPS:** Simulating CPU by Constructing and Composing essential Logical Components such as Adders, Registers, and MUXs using Verilog. [code] **(12/2021)**

- **Efficient Map Routing:** Graph Search Optimization for specific maps Applications that require frequent Queries beating The naive, classic Dijkstra's shortest path Complexity O(EV).[code] **(9/2021)**

Mentorship

- **Problem-Solving Training:** Completed 3 Levels under Coach Supervision. **(01/2019 - 04/2021)**
- **Open Source Society University:** self-taught curriculum in Computer Science, Studied Programming Languages, Computer Science Tools, and Software Design and Architecture**. (01/2024 - 06/2025)**

Competition Awards

- Running Kaggle Competition on White Blood Cell Classification under Severe Class Imbalance.
- Represented my College in **The International Collegiate Programming Contest (ECPC)**, 3 consecutive times in **(2020-2021-2022).**
- **Ranked #1** at Kaggle Machine Learning Completion**. (2022)**

Certification

- Huawei Certified HCIA-AI V3. certified. Sep 2022.
- Nvidia Deep Learning Workshop. certified. Sep 2022
- Egfwd Udacity Nanodegree cross-skilling. certified. Aug 2022 - Sep 2022
- ITI - AI Summer Training. certified Aug 2022 - Sep 2022\

Youtube Channel

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